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The future is not the present: Responsible pathways for forensic epigenetics☆

Nicole Palmour, Jessica Huang, Yann Joly *

Centre of Genomics and Policy, McGill University, 740 Ave. Dr. Penfield, Suite 5100, Montreal, Québec H3A 0G1, Canada

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ABSTRACT

Forensic epigenetics is emerging as a powerful extension of traditional forensic genetics, offering the capacity to infer age, lifestyle, and environmental exposures from epigenetic marks. Yet its promise is shadowed by significant ethical, legal, and social questions. This article analyzes the scientific foundations and practical applications of forensic epigenetic techniques while interrogating their implications for privacy, discrimination, and human rights. It argues that the promise of enhanced investigative capability must be balanced against risks of misuse, stigmatization, and function creep. Drawing on comparative perspectives in law and bioethics, the authors emphasize the importance of proportional governance frameworks that uphold transparency, accountability, and respect for persons. Suggestions are made for the responsible integration of epigenetic data in forensic contexts, if and when, it meets sufficiently rigorous standards.

1. Introduction

The rapid expansion of genomic and epigenomic science has transformed not only medicine and population health but also criminal investigation. Epigenetics, understood as heritable changes in gene expression that do not alter underlying DNA sequences, has opened new frontiers for understanding human identity, aging, and environmental interaction. Unlike traditional genetic markers, such as short tandem repeats (STR), which remain static across an individual's lifetime, epigenetic modifications reflect dynamic interactions between genes and the environment, providing a rich layer of contextual information. In recent years, forensic laboratories have begun investigating whether epigenetic modifications can be harnessed to infer cell type, biological age, behavioral patterns, and even individual identity with increasing precision [1–5].

Epigenetic modifications are mediated by a variety of molecular mechanisms that regulate the accessibility and activity of genes, allowing cells to respond dynamically to developmental cues and environmental stimuli. Core epigenetic mechanisms include DNA methylation, histone modifications, chromatin remodeling, and the regulatory functions of non-coding RNAs. In a forensic context, epigenetic information offers a novel layer of biological evidence that complements traditional genetic profiling. While STR analysis is limited to identity matching, epigenetic data may reveal phenotypic traits or environmental exposures.

This paper explores the evolving landscape of forensic epigenetics, examining the emerging applications and the critical challenges it poses to privacy, discrimination, and human rights. The discussion proceeds in four parts. The first reviews the emerging applications of forensic epigenetics, emphasizing the types of data generated and their uses. The second assesses critical challenges, involving biological complexity and technical limitations. The third examines the legal admissibility and evidentiary standards of introducing epigenetics in the courtroom. The fourth analyzes the human rights and ethical implications of collecting, analyzing, and storing epigenetic information for forensic purposes. Ultimately, we conclude that epigenetics is not yet fully prepared for widespread forensic application and recommend taking preparatory steps to ensure regulators and law enforcement are equipped for its responsible integration if and when the technology becomes sufficiently validated.

2. Emerging applications

Forensic epigenetics offers several promising applications that extend beyond traditional DNA profiling. Single-cell epigenomics enables analysis of DNA methylation and chromatin accessibility at the

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* Corresponding author.
E-mail addresses: nicole.palmour@mcgill.ca (N. Palmour), jia.huang2@mcgill.ca (J. Huang), yann.joly@mcgill.ca (Y. Joly).

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cellular level, improving identification in complex mixtures and providing context such as cell type and deposition sequence [6,7]. Epigenetic age clocks, based on methylation patterns at specific CpG sites, estimate chronological or biological age with mean errors of 3–5 years under optimal conditions [8,9]. When applied to degraded forensic samples, degradation refers primarily to DNA fragmentation and low-input templates typical of crime-scene traces, which constrain amplicon length and reduce assay sensitivity; additional chemical insults to cytosines (notably deamination and bisulfate-induced damage) can further bias methylation readouts and compound error if not controlled. Despite these challenges, adaptations using short-amplicon, targeted CpG panels show potential for narrowing suspect pools and aiding victim identification, though accuracy varies by tissue type and population background [10].

Tissue-specific methylation markers help determine the origin of biological traces (blood, semen, saliva) when morphological cues are absent, aiding crime scene reconstruction [11,12]. Similarly, lifestyle inference explores methylation signatures linked to smoking or environmental exposures, offering contextual clues but remaining probabilistic and ethically sensitive [10,11]. Initiatives like VISAGE integrate genetic and epigenetic predictors for appearance, age, and ancestry, while multi-omics approaches improve trait prediction accuracy [13,14]. Finally, twin differentiation, a longstanding challenge in forensic genetics, can be addressed through methylation and RNA expression differences, including circRNA profiles, which enable discrimination between monozygotic twins [15,16].

Advances in next-generation sequencing and high-throughput methylation arrays make large-scale epigenetic profiling feasible, but they also introduce challenges of data storage, interpretation, and governance. The ability to infer age, health, and environmental exposures from biological material transforms every trace sample into a repository of personal information, raising profound ethical and legal questions explored in subsequent sections. The capacity to extract information about age, health, or environmental exposure from biological material transforms every trace sample into a potential repository of personal and social meaning. The next section explores how this transformation unfolds in practice, highlighting both the investigative potential and ethical dilemmas of applying epigenetic data in criminal justice contexts.

3. Critical challenges

As forensic applications of epigenetics expand, they promise to enhance investigative capabilities in ways previously thought unattainable. However, this promise is tempered by a range of scientific, technological, legal, and ethical challenges. The contextual sensitivity and reversibility of epigenetic marks raise questions about reliability and reproducibility, while the lack of standardized assays and reference databases complicates implementation. Integrating epigenetic data into forensic workflows also demands careful attention to evidentiary standards, privacy, and potential misuse. Although still experimental, epigenetic analysis is reshaping debates on permissible evidence. By moving beyond identity confirmation to trait inference and biological context, it fills gaps left by conventional DNA profiling, but these same features heighten interpretive and ethical risks.

3.1. Biological complexity

The development and implementation of forensic epigenetic biomarkers face substantial biological challenges arising from four key characteristics: tissue and cellular specificity, environmental sensitivity, population-level variation, and temporal dynamics. Understanding these sources of variation is essential for robust method development, validation, and accurate interpretation of epigenetic evidence.

DNA methylation patterns vary significantly between tissue types, allowing differentiation between blood, saliva, semen, and other body fluids [12]. This necessitates either pre-identification of tissue source or development of tissue-independent protocols [11]. Within tissues, cell type-specific methylation signals further complicate interpretation, particularly for heterogeneous samples like whole blood [17]. Cell-type deconvolution methods (epigenome-wide association studies (EWAS)) that perform well in controlled laboratory settings often fail in forensic casework because forensic samples rarely meet the assumptions these algorithms depend on [18]. In EWAS, high-quality DNA, known cell types, and comprehensive reference methylation profiles enable accurate modeling, but forensic samples are typically degraded, environmentally altered, extremely low in quantity, and composed of unknown or complex mixtures of tissues for which no validated reference methylomes exist. Reference-based approaches break down when key discriminating CpGs are missing or when real samples contain unrepresented cell types, while reference-free methods misinterpret degradation, contamination, and environmental noise as biological variation because they assume that major components of variation arise from cell-type differences. Together, these constraints mean that current deconvolution tools cannot reliably interpret the heterogeneous, damaged, and unpredictable biological material typical of forensic evidence [18].

Epigenetic patterns are highly dynamic, influenced by sex, aging, environmental exposures, lifestyle factors, and disease states, making stable profiles difficult to capture. The interactions complicate interpretation: specific CpG sites show overlap between age- and smoking-associated methylation patterns, while chronic diseases may accelerate epigenetic aging [9]. The temporal variability adds further complexity: smoking-associated methylation changes revert to baseline within one year of quitting [19], and post-mortem degradation causes methylation changes, beyond 72 h [2], highlighting time sensitivity as the interval between reference and crime scene sample collection lengthens.

DNA methylation shows population-specificity at numerous CpG sites, with patterns observed in over one-third of all genes [20]. These differences arise as evolutionary adaptations to varying environmental and cultural influences and can be transgenerationally inherited. Consequently, forensic age prediction models show significantly reduced accuracy when applied beyond their training populations [21,22], requiring careful consideration of population ancestry in validation studies. These biological complexities are compounded by technical and analytical limitations in current measurement platforms, as discussed in the following section.

3.2. Technical limitations

The forensic validity of any molecular tool depends on reproducibility, sensitivity, and independence from confounding variables. Many epigenetic markers identified to date originate from small, demographically limited studies making predictive models vulnerable to bias when applied to diverse populations. Population-specific variation in methylation patterns can introduce systematic bias if predictive models are trained on non-representative cohorts. Moreover, sample degradation, common in forensic contexts, can distort methylation signals. These challenges are compounded by the absence of universally accepted protocols or accreditation standards comparable to those governing STR analysis. Current practice relies on diverse analytical platforms, reference datasets, and algorithms, creating significant inter-laboratory variability undermining evidentiary reliability [23,24]. For example, ISO/IEC 17025, which governs competence and validation in forensic laboratories, would need to incorporate requirements unique to epigenetic workflows, such as verifying bisulfate-conversion efficiency, establishing minimum CpG coverage thresholds, and validating bioinformatic pipelines used for methylation quantification and cell-type deconvolution [25]. ISO 18385, designed to minimize human DNA contamination, would require updates to address contamination risks specific to epigenetic testing, particularly the introduction of exogenous methylation signatures, degraded DNA artifacts, or environmental

factors that distort methylation patterns [26]. Additionally, forensic-specific guidelines modeled on SWGDAM or ENFSI would need to be expanded or newly developed to set standardized criteria for validating epigenetic markers, defining mixture-interpretation rules, setting population-appropriate reference datasets, and regulating computational models such as epigenetic clocks [27]. Together, with appropriate legal and ethical frameworks, these adapted standards would create a coherent framework for ensuring reliability, reproducibility, and admissibility of epigenetic evidence in forensic casework. Without robust validation and standardized protocols, premature implementation risks overinterpretation or misuse of data. Most epigenetic markers were discovered in medical or epidemiological studies, not in forensic populations, raising questions about reproducibility under real-world conditions [28]. Rigorous validation is essential to ensure accuracy and reliability in casework; the utility of these applications will hinge on their acceptance as legitimate evidence in court. Standardization and validation are therefore crucial to improve epigenetic forensic practice.

Integration of protocols into existing forensic laboratories demands compatibility with current systems, yet high costs and equipment requirements remain barriers, particularly in resource-limited settings. Standardized protocols for sample collection, processing, and analysis across jurisdictions are critical [29]. Accurate interpretation demands attention to biological and technical factors such as tissue type, aging, lifestyle, and disease. Longitudinal and post-mortem studies are needed to assess methylation stability over time, but such research is still lacking [23,24]. Developing robust reference databases, similar to those in forensic genetics, is necessary to provide population-level frequencies for epigenetic markers [10,30].

Dedicated forensic epigenetic databases, including ante- and post-mortem methylation and histone data, will be required for practical application [2]. Validation and reporting guidelines remain another critical gap. Rigorous studies are needed to establish accuracy, sensitivity, and specificity for forensic contexts. Clear reporting standards, such as whether to disclose individual methylation levels or age ranges, must be developed alongside user-friendly analytical tools [31]. Finally, successful implementation will depend on public engagement, transparent governance, and specialized training for forensic practitioners. Global collaboration, data sharing, and cost-benefit analyses will be essential to ensure ethical and effective integration of epigenetic methods into forensic practice.

4. Legal framework

This section will analyze how the preceding challenges intersect with human rights law, privacy protection and outlines frameworks for responsible governance. There is a strong interest from police authorities in being able to use the latest technology for crime prevention. While this is a laudable intent, using information rich data without first designing proper legal and ethical frameworks can carry risks for the population. Because epigenetic information can reveal not only identity but also health status, environmental exposures, and behavioural history, its use in criminal justice engages privacy, autonomy, and equality in ways not fully anticipated by current law. While reliable epigenetic testing may assist police in identifying suspects if collected lawfully, this does not mean results are automatically admissible as evidence in court.

4.1. Evidentiary standards and admissibility

The introduction of epigenetic data as evidence in a criminal trial could be useful to distinguish identical twins from each other [32], or as additional evidence to support proof collected via more traditional investigative techniques. However, the production of epigenetic evidence in court could be challenged using two distinct legal arguments. While these arguments are found across jurisdictions, their concrete applications may differ substantially from one country to another. It could be argued that 1) the introduction of the epigenetic evidence is breaching the fundamental rights and freedoms of the accused person, or 2) epigenetics test results are insufficiently reliable or relevant to be introduced as evidence in criminal proceedings. In addition to overcoming those two objections, it will be important for prosecutors to demonstrate the integrity of the chain of evidence from the crime scene to the lab, and the courtroom.

4.1.1. Reliability and relevance

Courts will likely approach epigenetics evidence as investigative intelligence rather than standalone identity proof until there are consensus protocols, inter-laboratory validation, and transparent error models that handle tissue heterogeneity, ancestry effects, and degradation. Epigenetics results should be framed as probabilistic and contextual, similar to other pattern evidence, accompanied by uncertainty intervals and conservative limiting instructions [33]. Responsible reporting requires: (i) disclosure of the model, training population, and error margins; (ii) identification of known confounders (disease, smoking, stress, post-mortem interval, population ancestry); (iii) avoidance of causal or deterministic language; (iv) documentation of chain-of-custody and analytical protocols; and (v) limiting scope to validated applications, resisting lifestyle or health inferences unless specifically authorised by a search warrant.

Successfully linking epigenetic markers from a crime scene to an individual does not prove guilt. It may indicate presence at the scene or shared traits with the suspect, but only when supported by other evidence, such as standard genetic testing.

4.2. Human rights foundations

International human rights law provides a foundational lens for evaluating the governance of forensic epigenetics. The rights to privacy, bodily integrity, and non-discrimination, enshrined in instruments such as the *Universal Declaration of Human Rights and the International Covenant on Civil and Political Rights*, require that state intrusions meet principles of proportionality and necessity [34,35]. While collecting biological material for law enforcement may be permissible under strict conditions, repurposing it to infer sensitive personal information, such as lifestyle, health status, or ancestry, risks exceeding these limits.

The Council of Europe's *Convention on Human Rights and Biomedicine* (Oviedo Convention) and the EU *General Data Protection Regulation* (GDPR) emphasize that processing biological data must be "adequate, relevant, and limited to what is necessary" [36,37]. These core privacy principles are also recognized in the data privacy legislation of numerous countries outside the EU. However, in Europe the Prüm Framework (Prüm Convention + Decisions) creates a parallel security-driven legal basis for cross-border genetic data exchange among law-enforcement authorities [38]. Prüm is designed to support serious crime and terrorism investigations, not medical or research uses. It therefore recontextualizes DNA as law-enforcement biometric data, not as sensitive health-related genomic data. Epigenetic profiling for investigative purposes challenges these principles, as the data collected often extend far beyond identification needs. While it is uncertain that epigenetic data could be shared under the Prüm framework as this would require epigenetic data to be legally characterized as DNA profile, future governance mechanisms should ensure that forensic epigenetic applications are narrowly tailored to legitimate objectives and subject to independent oversight.

Proportionality becomes a crucial guiding principle in this context. The intrusion on personal privacy must be commensurate with the public interest at stake. When the science becomes sufficiently robust, forensic epigenetic analyses should be restricted to cases of serious crime and applied only when less intrusive methods are unavailable. Routine or speculative uses, such as population screening or databasing epigenetic profiles, would violate proportionality and risk normalizing molecular surveillance. Consent further complicates this picture: individuals compelled to provide samples cannot meaningfully consent

to analyses that reveal information they never agreed to disclose. Legal authorizations are generally narrowly framed to permit collection for specific purposes, not open-ended exploration of incidental findings.

Beyond its potential utility, epigenetic evidence in criminal proceedings poses risks that could undermine fundamental rights and freedoms of the accused. Unlike traditional DNA profiling, epigenetic markers can reveal highly sensitive information, including health status, lifestyle habits, environmental exposures, and even predispositions to disease, extending far beyond identity verification [10,39]. Such disclosures implicate privacy, autonomy, and the right not to know, as recognized in rights-based frameworks [40,41]. Moreover, epigenetic data may have transgenerational implications, potentially affecting family members and creating risks of discrimination or stigmatization [42,43]. These characteristics challenge the proportionality principle in criminal law, which requires balancing investigative benefits against infringements on liberty and dignity [44].

Additional concerns include the lack of scientific validation and standardization (see below), which undermines reliability under evidentiary standards such as *Daubert* [45,46], and the potential for unlawful overreach if epigenetic data is used beyond identification purposes [47]. Legal scholars caution that, given the current scientific limitations, epigenetic findings should not be used to alter doctrines of culpability or punishment, as premature integration could erode fairness and due process [48].

4.3. Scientific reliability: the Daubert standard

In most jurisdictions, because of the presumption of innocence, the prosecution bears the burden of proving guilt and may seek to introduce scientific evidence through expert testimony. The "Daubert Standard", established by the U.S. Supreme Court [45], provides criteria for assessing scientific reliability and relevance before presentation to a jury. While not legally binding outside of the U.S., Daubert has become the de facto international standard for evaluating novel scientific evidence [49–51]

The Daubert criteria require consideration of the following:

- Has the technique been tested in actual field conditions (beyond laboratory settings)?
- Has the technique been subject to peer review and publication?
- What is the known or potential rate of error?
- Are there established standards or guidelines for ensuring the proper operation of this technique?
- Has the technique been widely accepted within the relevant scientific community?

For epigenetic age estimation and related methylation assays, peer-reviewed literature and validation studies increasingly satisfy the first two criteria. However, error rates and population transferability remain critical stress points. Epigenetic age clocks demonstrate mean absolute errors of 3–5 years under optimal study conditions, but performance declines with sample degradation, tissue heterogeneity, population ancestry differences, and confounding factors like disease and stress (as outlined previously in Section 3).

It is important to recognize that the Daubert Standard was developed in a civil proceeding and that there is still some fluctuation between courts and jurisdictions on the way it is applied in criminal proceedings. Risinger [52] and Neufeld [53] are of the opinion that *Daubert* motions are rarely made by criminal defendants and that the few such challenges that have been made did not fare well. Yet, commenting on the specific context of DNA phenotyping evidence, Hodge and Meehan [46] conclude that the lack of peer-reviewed studies and failure of the technique to gain general acceptance in the scientific community could render DNA phenotyping inadmissible under *Daubert*.

4.4. Legal frameworks for using epigenetic age estimators in forensic investigations

Forensic DNA phenotyping (FDP) refers to a set of techniques used to analyze biological material collected from crime scenes to infer an individual's biogeographical ancestry and externally visible physical characteristics, such as eye, hair, and skin colour. FDP has been used in several high-profile cases in different jurisdictions to generate investigative intelligence and provide leads relevant to specific suspects or populations of interest. The introduction of FDP marks a shift in how forensic genetics is used within the criminal justice system. First, it reorients forensic science from producing evidence for courtroom identification toward generating investigative intelligence that can inform police decision-making. Second, by identifying groups of potential suspects who share common ancestral or phenotypic characteristics, FDP represents a movement away from strict individualization and toward collectivization in forensic practice [54].

While most data used for DNA phenotyping are currently not epigenetic data, a common practice in with this technology is to use DNA methylation or telomere clocks, to estimate the age of individuals [55,56]. The use of this technology for police investigation is particularly preoccupying considering the lack of precision of current epigenetic clocks [21]. Given the perceived broad potential of epigenetics to reconcile the biophysical with the environment it is quite possible that additional epigenetic data will be used in the future to document other physical characteristics.

While some argue that forensic phenotyping does not infringe dignity or privacy because it focuses on externally visible traits [57], this perspective is overly simplistic. Traits such as ancestry and age are not merely physical descriptors; they constitute sensitive personal information that can reveal social identity and life history. In the context of digitized policing, these data points contribute to systemic privacy risks, reinforcing power asymmetries between authorities and individuals [58]. If this investigative technique is carried out without ethical foresight and absent necessary good ethical standards, the situation can become rife for human right infringements and racist practices [43]. Using ancestry information carries significant risks, including potential discrimination against minority groups, infringements on privacy and autonomy, and misinterpretation of forensic DNA phenotyping results by various stakeholders [43,58].

Outside research settings, the use of epigenetic data in forensic investigations and criminal proceedings has so far been limited. Thus, the complexity of epigenetics and the need for additional validation studies, likely explain the absence of criminal provisions that would explicitly regulate the use of epigenetic markers for criminal investigations. DNA phenotyping laws have been adopted in a few countries in Europe (see Table 1 below) and two of these laws explicitly allow the use of DNA methylation data to estimate the age of a person from a sample found at a crime scene or, are formulated in a manner that seem sufficiently broad, to permit it.

Table 1
Countries that explicitly regulate forensic DNA phenotyping.

Country	Legislation	Phenotyping for age estimation
Germany (Bavaria)	*Bavarian Police Duties Act 2018*[67]	Yes
The Netherlands	*Dutch Code of Criminal Procedure 2012* [68]	No
Slovakia	*the Act on the use of DNA analysis to identify individuals 2002*[69]	No
Switzerland	*Federal Act on the Use of DNA Profiles in Criminal Proceedings and for Identifying Unidentified or Missing Persons 2003*[70]	Yes

Countries with explicit legislation on forensic DNA phenotyping and its application to age estimation

The absence of comprehensive laws on the use of epigenetics markers for forensic investigation does not preclude police authorities from attempting to use this science to derive information from a sample found at a crime scene [59]. In many jurisdictions, such a sample would legally be considered abandoned material and therefore free for use by law enforcement authorities. In Canadian law for criminal evidence to be acceptable in court it must be shown to be admissible, material, and relevant. To be admissible, evidence must have at least some probative value that outweighs any prejudicial effect; when applied to lifestyle-related epigenetic markers (e.g., methylation signatures associated with smoking or alcohol use), this balance tips unfavorably because such markers rarely speak to a fact in issue (identity, presence, mechanism) and instead offer population-level, probabilistic associations that are highly sensitive to environmental and temporal confounding [60]. Further, their relevance to the case tends to be marginal, while the legislative risks, including privacy violations, the danger of character inferences, and the likelihood of misleading the trier of fact through an aura of unwarranted scientific certainty, are substantial [61]. Similar rules can be found in other jurisdictions [61]. Given the current dearth of good practices, professional standards, etc. for using epigenetics evidence in forensic studies and given some of the known scientific limitations, such evidence would likely have low probative value and fail admissibility standards in court [2]. The legal and ethical impact on public perception of the criminal justice system of using insufficiently validated epigenetic evidence, such as ancestry prediction, should not be underestimated [62]. In 2022, in Canada, Edmonton police employed forensic DNA phenotyping (but not epigenetic data) to create a suspect sketch in an unresolved sexual assault case. After the sketch was circulated, community organizations and legal experts criticized the approach as constituting racial profiling and for other ethical shortcomings. The police subsequently withdrew the image and issued a public apology [63]. This incident underscores the need for the police to issue clear publicly available guidelines and carry out public consultation before implementing such far-reaching technologies.

In the U.S., forensic DNA phenotyping is used in at least 41 states [64]. However, there is no law at the federal or state level explicitly authorising the use of this information thus its legal status rests on general evidentiary principles and broad interpretations of investigative authority. This ambiguity becomes more acute when considering epigenetic markers, since unlike standard SNP-based phenotyping, many methylation markers do not merely predict externally visible traits but instead may signal lifestyle exposures and underlying health conditions. However, statutory restrictions such as the Texas Government Code § 411.143(d), which expressly forbids using the DNA database for information about human physical traits or disease predisposition unless that trait/predisposition is required for one of the permissible purposes [65]. The relevant section of this regulation then describes the permissible purposes as investigation or prosecution of offences; identification and exclusion of suspects; human remains or missing persons' work; population statistics and forensic validation [65]. This broad exception means that DNA phenotyping is acceptable by law in Texas for criminal investigations and prosecution purposes. Epigenetic loci such as PEAR1 and SLC7A11 illustrate the problem: methylation changes at these sites can reflect alcohol consumption patterns, but they simultaneously correlate with cardiovascular risk, platelet function, oxidative-stress pathways, and broader metabolic health. In practice, this means a single epigenetic measurement can blur the legally significant line between permissible investigative inference (e.g., inferring a lifestyle exposure relevant to a suspect profile) and prohibited medical inference (e.g., revealing disease susceptibility or health status).

There are few domestic laws that explicitly regulate DNA phenotyping, and those laws are currently too distinct from one another to promote the emergence of a harmonized international standard [44].

Four European countries, Switzerland, the Netherlands, Germany and Slovakia and some U.S. States explicitly regulate the use of forensic DNA phenotyping in their legislation [64,66]. Table 1 below presents the legislation of these countries and comments on whether it would allow, implicitly or explicitly, the use of DNA methylation analysis for age assessment. Other European countries including the United Kingdom, France, Poland, Czech Republic, Sweden, Hungary, Austria, Spain have used FDP in criminal investigations relying on general DNA statutes or police powers rather than an explicit phenotyping clause.

Outside of North America and Europe, there is currently no evidence that DNA phenotyping is used by police authorities. However, China is known for its extensive police DNA practice, a very broad legal framework and documented scientific research on DNA phenotyping [71].

4.4.1. *International policies on forensic epigenetics*

Although several articles have explored the potential of epigenetics in forensic investigations, stakeholders remain focused on gathering scientific evidence and addressing implementation challenges. As a result, only a few policy standards currently exist, and these address forensic epigenetics in very general terms. In Europe, there are general guidelines on forensic DNA and human rights – such as Council of Europe Recommendation R(92)1 and subsequent EU guidance limiting forensic DNA to non-coding, non-health-related loci that are relevant to this practice. Internationally, the Human Epigenome Consortium (IHEC), founded in 2010, aimed to generate human epigenomes from diverse cell types and maintained an active bioethics workgroup from 2012 to 2024. While some of its guidelines touched on issues relevant to forensic applications, such as privacy, communication of results, and research involving vulnerable populations, the focus was primarily on medical epigenetics rather than forensic use [72].

Two European reports have issued some relevant recommendations; the 2007 Nuffield Council's *Report on the forensic use of bioinformation: ethical issues* which is general in scope, and the VISAGE Consortium *Recommendations to address the ethical and societal challenges of forensic DNA phenotyping* [73,74]. While the Nuffield Council's Report remains influential, it was adopted at a time where epigenetics and DNA phenotyping were not widespread [73]. It addresses the proportionality and necessity criteria applicable to forensic DNA use; strict limitations on expanding from identification profiles to more informative analyses; risks of discrimination, function creep, and stigmatization for particular groups; the need for independent oversight and public debate for the introduction of new technologies.

The VISAGE *Recommendation*s, while more recent and specific in scope, has been produced under the mandate of an organization of stakeholders that include several police related organisations and whose objectives are "to develop, validate and implement a set of new prototype DNA-based tools to allow for appearance, age, and biogeographical ancestry (BGA) prediction testing of unknown perpetrators from various crime scene traces in an efficient way" [74]. While the two co-authors are academics, the greater context of the publication explains why the Report focusses on ethically responsible implementation and does not question if/when the technology should be used at all. Key recommendations include limiting FDP to a narrow set of traits (appearance, biogeographical ancestry, age) with good scientific validity; prohibiting health-related, behavioural, or personality inference; using FDP only for serious crime, and ideally as a tool of last resort; robust validation, documentation, and transparency in reporting and paying special attention to racialization and discrimination risks, especially around ancestry prediction [74]. Finally, the U.S. National Institute of Standards and Technology (NIST) recently issued a standard addressing human factors in forensic DNA interpretation [75]. Although focused on STR analysis, its 44 recommendations to reduce human error could be adapted for epigenetic applications once validation is achieved.

Other international forensic ethics guidelines include *Ethical publication of research on genetics and genomics of biological material: guidelines and recommendations* [76], jointly published by *Forensic Science International: Genetics* and *Forensic Science International*. These guidelines, along with more recent editorial standards [77], provide general principles for ethical forensic genetic research that can be extended to

epigenetics. However, they remain broad and offer little specific guidance for forensic applications. Specific epigenetic considerations were introduced by Naue and Lee [78], who called for recommendations on publishing DNA methylation results to standardize research practices. They emphasized that methylation research and its forensic applications are still in their infancy compared to other areas of forensic genetics.

5. Ethical concerns

Biometric technologies, including fingerprints, facial recognition, DNA, and emerging epigenetic methods, are transforming forensic science but raise profound ethical challenges. These identifiers are inherently personal and sensitive, and their collection often occurs without consent, creating tensions between public safety and individual rights. While traditional DNA profiling sought to minimize health-related implications, epigenetic approaches introduce additional layers of complexity by revealing traits linked to health, lifestyle, and aging [1–5]. For example, methylation patterns at CpG sites in genes such as *AHRR* and *F2RL3* can indicate smoking exposure, while signatures in *PEAR1* and *SLC7A11* suggest chronic alcohol use. However, these methylation changes reflect probabilistic associations, not definitive evidence of behaviour. It is therefore essential that reports avoid deterministic statements, such as asserting that an individual is a smoker, and instead communicate findings in conditional terms (e.g., "the methylation profile is consistent with an increased likelihood of prior smoking"). Clear probabilistic framing is necessary to prevent over-interpretation and to ensure that epigenetic results are understood within their scientific limits. Epigenetic clocks based on methylation profiles estimate chronological age within 3–5 years, and deviations may reveal accelerated aging due to stress or disease. Other markers can reflect environmental exposures, such as air pollution, or even predispositions to conditions like cancer or diabetes, adding sensitive health and lifestyle dimensions to forensic evidence. Together, these technologies implicate privacy, autonomy, fairness, and public trust, underscoring the need for robust ethical frameworks and transparent governance to ensure responsible innovation [41,79].

5.1. Generic ethical concerns with all biometric identifiers

The use of biometric identifiers in criminal contexts presents a myriad of ethical challenges that warrant careful consideration. These inherently personal and sensitive biometric identifiers raise significant privacy concerns when collected or stored improperly [80]. In criminal investigations, their involuntary collection often limits autonomy and consent, infringing on personal rights [81]. Additionally, the accuracy and reliability of biometric systems are not infallible; these technologies can produce false positives or negatives and exhibit biases against certain demographic groups, potentially leading to miscarriages of justice [82,83]. Security risks are a critical concern, as biometric data is a prime target for cyberattacks, enabling identity theft and other malicious activities [84]. Moreover, the potential for misuse of biometric data for mass surveillance poses a threat to civil liberties, particularly in authoritarian regimes where such data could be exploited to suppress dissent [81]. Finally, lack of transparency and accountability in data handling undermines public trust, highlighting the need for robust ethical frameworks to govern forensic applications [80,84].

5.2. Shared issues with genetics and epigenetics

5.2.1. Privacy, bias, and data protection in forensic epigenetics technology

Epigenetic testing introduces ethical challenges that extend beyond individual privacy to the structural properties of the data and the models on which forensic inference relies. Unlike traditional forensic STR profiling, which is limited to identity verification, epigenetic markers are designed to produce probabilistic estimates of traits, age, or environmental exposures by reference to population-level patterns in methylation data [85]. As a result, epigenetic inference is inherently dependent on reference datasets that implicitly encode assumptions about biological normality, exposure histories and population structure.

This reliance introduces systemic sources of bias at the point of knowledge production. Reference datasets used to train epigenetic prediction models often underrepresent minority groups or fail to account for the influence of environmental, socioeconomic or chronic stress influencing methylation patterns [23,41]. These factors are biologically meaningful confounders that shape epigenetic variation and complicate the validation of predictive outputs. Consequently, predictions of age, lifestyle, or ancestry are subject to uneven error distributions that correlate with social marginalization rather than individual conduct [43].

Beyond bias, the epistemic scope of epigenetic data raises concerns about proportionality. Inferring lifestyle or environmental exposures from forensic samples risks normalizing molecular surveillance, shifting the focus from crime-specific evidence to broader behavioral or social traits [23]. This expansion challenges foundational principles of necessity and minimal intrusion, as individuals become implicated not for direct evidence of wrongdoing but for belonging to biologically inferred categories. Addressing these concerns requires governance frameworks that explicitly prohibit the use of epigenetic predictions for speculative or population-based screening, mandate transparency in algorithmic design, and enforce cultural sensitivity in interpretation [41,43].

5.3. Specific ethical challenges with epigenetic tests and data

5.3.1. Accuracy, interpretation, and incidental findings

Epigenetic markers are dynamic, making profiles unstable and sometimes unreliable for identification. In the forensic context, multiple external factors influence the accuracy and reliability of epigenetic data [23]. Reliability also depends on the evidence that links a variant to disease and the magnitude of associated risk [40]. These considerations become ethically complex if disease-associated profiles are used for identification, implicating the "right not to know," as individuals may not wish to learn of a genetic or epigenetic predisposition uncovered during forensic analysis. If actionable health risks are detected, questions may arise about whether such findings should be disclosed, particularly when preventive measures or treatments are available. However, forensic scientists are mandated to analyze and interpret biological material for evidentiary purposes, not to provide medical counselling, and investigators lack both the expertise and neutrality required for sensitive health-related communication. For these reasons, consultation with outside clinical experts who can determine whether disclosure is legally justified, medically relevant, and consistent with the individual's privacy rights may be in order.

5.3.2. Privacy, bias, and group-level implications

While section 5.2.2 examined the epistemic and data-level sources of uncertainty and bias in forensic epigenetics, this section considers how those limitations are amplified through institutional interpretation and use. Forensic age estimation and phenotyping do not operate in a social vacuum; rather, they are embedded within policing and judicial contexts that shape how probabilistic information is understood and acted upon.

A central concern arises from the treatment of probabilistic outputs as functionally definitive within investigative decision-making practices. Although epigenetic predictions generate likelihoods and confidence intervals rather than individual identification, these outputs are often operationalized as tools for narrowing suspect pools. In practice, this produces biologically inferred collectives: groups defined by shared predicted traits such age range, as eye, hair, and skin color, or ancestry, rather than individualized suspects [43]. These collectives frequently align with historically racialized or socially salient categories, creating pathways for stigmatization and discriminatory profiling, reintroducing structural inequalities through ostensibly neutral forms of biological inference.

The collectivization of suspicion has distinct ethical implications that are not reducible to technical bias alone. When investigative cultures and institutional incentives encourage the aggregation of probabilistic indicators into actionable intelligence, error and uncertainty are redistributed across communities rather than individuals [41]. This dynamic can intensify existing inequities in policing, disrupt family and community life, and erode trust in law enforcement, particularly in jurisdictions lacking clear oversight or constraints on the interpretation of forensic prediction [43].

These group-level implications underscore the need for governance mechanisms that extend beyond technical validation. Ethical oversight must address not only how epigenetic models are built, but how their outputs are translated into investigative action. Without safeguards that limit population-based screening, mandate contextualized interpretation, and recognize the social consequences of collective inference, forensic epigenetics risks undermining fairness and legitimacy despite its ostensible evidentiary neutrality [41,43].

5.3.3. Public knowledge and education gaps

Epigenetics remains poorly understood by the general population [86,87]. The complexity of epigenetic mechanisms, such as DNA methylation and histone modification, poses a major barrier to widespread understanding. These processes involve intricate biochemical interactions that are difficult to convey through mainstream media or general educational materials lacking nuanced discussion [88,89]. Consequently, many individuals lack a basic grasp of how environmental factors such as diet, stress, and exposure to toxins can influence gene expression and, in turn, health outcomes, further complicating public perceptions of how these technologies intersect with health, family, autonomy, privacy, security, and community.

5.3.4. Communication, transparency, and public trust

Previous research indicates that epigenetic findings are often overstated in both scientific publications and mainstream media, contributing to public hype and misconceptions about complex concepts [87,88]. In the forensic context, clear communication about the actual capabilities of forensic epigenetic technologies, their case-specific utility, reliability outside the laboratory, and the nature of the data they produce, has been strongly recommended [41,79]. Transparency regarding ethical considerations, limitations, accreditation processes, and regulatory and peer oversight can promote accountability, build legitimacy, foster public trust, and support responsible innovation [41,79,90].

5.3.5. Global disparities in scientific capacity

Global disparities in scientific capacity create significant variation in technical capacity and, standards for consent, data protection, and oversight. Joly [91] notes that while many countries in the Global North have established forensic DNA databases, only a limited number of jurisdictions in the Global South have developed similar infrastructures, raising the risk that low-resource settings may adopt imported technologies without implementing parallel ethical or legal frameworks. Such conditions heighten the risk of "ethics dumping", where ethically sensitive forensic practices, including emerging epigenetic tools, are employed in contexts lacking adequate regulatory protections. To mitigate these risks, international collaboration and harmonized governance standards are essential.

5.3.6. Accountability and oversight

Effective governance requires mechanisms that ensure transparency, accountability, and public trust [92]. Independent oversight committees should review the scientific validity and ethical aspects of proposed forensic epigenetic applications [93,94]. These bodies must have authority to review protocols, audit databases, and investigate complaints. Public engagement is equally essential. The history of forensic DNA demonstrates that legitimacy depends not only on scientific validity and accuracy, but also on social acceptability, the perception that a technology serves the public good and respects individual rights [95,96]. Clear communication about the capabilities and limitations of epigenetic analysis can help prevent both unrealistic expectations and undue alarm. At the international level, coordination between various parties interested in epigenetics data such as health researchers, police authority, bioethics councils, data protection authorities, and forensic agencies could promote better informed and more harmonized standards [97]. Initiatives such as the Global Alliance for Genomics and Health (GA4GH) Bioethics Committee, though limited in time, provide a useful organisational model for future governance organizations in the field of epigenetics [98].

5.3.7. Risk of function creep

The history of forensic genetics illustrates how technologies developed for narrow purposes can gradually expand beyond their original scope [99,100]. Databases established for serious crimes have widened to include minor offenses and even non-convicted, suspected, individuals [99]. Epigenetic data, with their capacity to reveal health, environment, or ancestry, are particularly vulnerable to such expansion [2,101]. They could be repurposed for predictive policing, oppression of minority groups, insurance assessment, immigration control, or other discriminatory practices [102,103]. Unlike STR profiles, which are relatively information-poor, methylation data encode multifaceted biological signals. Cross-referencing epigenetic profiles with medical or research databases could yield sensitive insights into an individual's lifestyle or disease susceptibility [104,105]. Preventing function creep will require explicit statutory safeguards and independent oversight of data retention, access, and cross-sector sharing [94].

6. Conclusion

Forensic epigenetics sits at a rapidly evolving crossroads between molecular biology, data science, and criminal investigation. Its scientific promise is genuine: age estimation, tissue identification, and contextual inference could one day refine investigative leads and complement existing forensic methods. Yet at present, the field is scientifically immature and ethically unprepared for operational deployment. The temptation to translate a promising research tool into an investigative instrument must be tempered by a clear understanding that epigenetic signatures are dynamic, context-dependent, and not yet sufficiently validated for reliable, fair, or interpretable use in criminal justice settings.

Given this, forensic epigenetics should not presently be incorporated into the investigative toolbox. To ensure readiness for future developments, police agencies, legislators, and forensic authorities should prioritize strategic preparation and capacity-building, positioning themselves to adopt the technology responsibly as its reliability and utility become established. This preparatory phase should prioritize meaningful engagement with the public and relevant stakeholders, ensuring that diverse communities, including but not limited to victims' families, have opportunities to understand the technology, express concerns, and shape its potential governance [106]. It should also include a clear, transparent roadmap outlining proposed applications, anticipated benefits, and the safeguards required to manage risks. In addition, preparatory work should involve targeted education of investigators, judges, policymakers, and the public; systematic mapping of foreseeable harms, such as discrimination, function creep, and surveillance drift; and the co-design of transparent, proportionate, rights-respecting governance and oversight frameworks that could be implemented if, and only if, epigenetic techniques ultimately meet rigorous scientific and legal thresholds.

Any future regulatory framework must go beyond laboratory validation to address interpretive uncertainty, contextual misuse, and equity concerns, and must be grounded in human rights, proportionality, necessity, accountability, and data minimisation. Cross-linkage with medical, research, commercial or administrative datasets should be

categorically prohibited without explicit consent and narrowly circumscribed judicial authorisation. Independent, external oversight, not merely internal policy, will be essential.

Ultimately, the current question is not how quickly forensic epigenetics can be introduced, but whether it should be introduced at all under present conditions. As the first references to epigenetic markers begin to appear in national legislative debates, there is a narrow window to shape governance before technological enthusiasm hardens into practice. Used responsibly and transparently, epigenetics could one day contribute to just investigations; used prematurely, it risks becoming a new vector of error, inequity, and intrusion. The wisest path forward is therefore measured anticipation rather than early adoption, a deliberate investment in ethical, legal, and social readiness commensurate with the sensitivity of the biological information at stake.

CRediT authorship contribution statement

Yann Joly: Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Jessica Huang:** Writing – review & editing. **Nicole Palmour:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation.

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Declaration of Competing Interest

The authors have no financial conflicts of interest to declare.

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